MILITARY SPECIFICATION

TRANSPARENCY, ANIMATED, M16A1 RIFLE LOCKING, UNLOCKING AND EJECTING MECHANISMS

This specification has been approved by the Naval Training Device Center, Department of the Navy. The technical requirements for procuring the animated transparency described herein shall consist of this document and the issue in effect of Specification MIL-T-82391.

1. SCOPE

1.1 This specification covers the requirements for an animated transparency depicting the locking, unlocking and ejecting mechanisms of the 5.56-MM M16A1 rifle.

1.1.1 This item has been designated by the Naval Training Device Center as Device 3F68/1B for Navy procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-82391	Transparencies, Animated: Design and Development of, General Specification for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

FSC 6910

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3. REQUIREMENTS

3.1 Material.- Except as specified in 3.1.1 and 3.1.2, materials used shall be in accordance with the material requirements of MIL-T-82391.

3.1.1 Metals.- All metal parts and fastenings, excluding springs, shall be of corrosion resistant steel, class 2 chromium plated metal in accordance with QQ-C-320 or Type II anodized aluminum in accordance with MIL-A-8625.

3.1.2 Wood.- Wood used shall be select grade hardwood suitable for a painted finish. The wood shall be properly seasoned, free from checks and splits and shall not be warped.

3.1.2.1 Glue.- Glue used for joining wood surfaces shall be a type not affected by moisture, lacquer, paint or sealers and shall be applied only to bare wood surfaces.

3.2 Design.- The animated transparency shall be designed to simulate, in detail, the operation of the locking, unlocking and ejecting mechanisms in the firing cycle of the M16A1 rifle. The design shall be in accordance with the design requirements of MIL-T-82391 and as specified in paragraph 3.2.1 of this specification.

3.2.1 Design details.- The animated transparency shall be designed for use in demonstrating the operation of the bolt assembly, the bolt carrier and the bolt cam pin in locking and unlocking the bolt and extracting and ejecting the empty cartridge.

3.2.1.1 The following major components shall be depicted:

(a) A small projectable pictorial view of the entire rifle

(b) Ejector port cover

(c) Partial barrel assembly

(d) Bolt carrier

(e) Bolt assembly

(f) Bolt cam

(g) Cartridge

(h) Firing pin

(i) Upper receiver assembly.

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3.2.1.2 The pictorial view of the rifle, as required by 3.2.1.1 (a) shall be gray in color and that portion of the rifle demonstrated by the animated transparency shall be amber in color. The nomenclature "M16A1 RIFLE" shall be included. The view shall be provided with an opaque flap.

3.2.1.3 All operative and stationary parts of all components necessary to demonstrate the operation of the locking, unlocking, extracting and ejecting mechanisms shall be depicted.

3.2.1.4 Means shall be provided for depicting the rotation of the bolt assembly, caused by the action of the bolt cam within the bolt carrier cam track, in both the locking and the unlocking sequences.

3.2.1.5 Means shall be provided for indicating the opening of the ejector port cover by the movement of the bolt carrier.

3.2.1.6 Means shall be provided for indicating the cartridge being held by the extractor until the rearward movement of the bolt carrier exposes the cartridge to the ejector port.

3.2.1.7 Means shall be provided for clearly indicating the function of the ejector at the time the cartridge is aligned with the ejector port.

3.2.1.8 Means shall be provided for rechambering and relocking the same cartridge repeatedly.

3.3 Reliability.- The animated transparency shall meet the reliability requirements of MIL-T-82391.

3.4 Details of components.- Except as specified in 3.4.1 and 3.4.2, the details of components of the animated transparency shall be in accordance with the details of components requirements of MIL-T-82391.

3.4.1 Baseplate markings.- Information for orientation of the animated transparency, identification of controls or for warning notes shall be engraved with .250 inch high letters and filled with a color (black or white) contrasting with the background, by application of dry transfer lettering, by silk screening or by fluid lettering devices. Markings not engraved shall be protected by transparent sheet plastic.

3.4.2 Storage/carrying case.- The storage/carrying case for the animated transparency shall be in accordance with the storage/carrying case requirements of MIL-T-82391 except that the material for the top and bottom of the case shall be 1/4" plywood.

3.5 Construction.- Construction shall be in accordance with the construction requirements of MIL-T-82391.

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3.6 Performance.- Performance shall be in accordance with the performance requirements of MIL-T-82391.

3.7 Finish and color.- Finish and color shall be in accordance with the finish and color requirements of MIL-T-82391.

3.8 Nameplates or product markings.- Nameplates or product markings shall be in accordance with the nameplates or product markings of MIL-T-82391.

3.9 Workmanship.- Workmanship shall be in accordance with the workmanship requirements of MIL-T-82391.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspections, examinations and tests.- Inspections, examinations and tests shall be in accordance with the quality assurance provisions of MIL-T-82391.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking shall be in accordance with the preparation for delivery requirements of MIL-T-82391.

6. NOTES

6.1 The notes of MIL-T-82391 are applicable.

CUSTODIAN:
Navy-TD

PROCURING ACTIVITY:
Navy-TD
Project No. 6910-N298

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM, APR 62 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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